

Lift Predictive Maintenance Programme - Maintenance Record FY2024

Southbank Office Tower · Office · Melbourne

Document ID	OFF-04
Category	Vertical Transportation
Classification	Maintenance
Date	2025-01
Prepared by	Facilities Management, Australia Office Portfolio

1. Baseline (FY2023)

Southbank Office Tower has 12 passenger lifts, installed in 2008, serving 34 floors. In FY2023 the lifts recorded 58 breakdowns and 7 passenger entrapments, and lift availability was 98.7%. Maintenance was time-based: the lift contractor visited each lift monthly regardless of its condition, and most faults were only discovered when a lift stopped.

Fault category	Share of breakdowns
Door operator, door locks and sills	61%
Drive and traction motor	18%
Controller and levelling	14%
Other	7%

2. Impact on tenants

Breakdowns concentrated in the morning peak caused long waits in the lobby, and each entrapment triggered an incident report and a tenant escalation. Lift performance was the second most common complaint in the FY2023 tenant survey.

3. Condition-monitoring programme

- IoT sensors retrofitted on each lift: door cycle counter, door motor current, car vibration (accelerometer), levelling accuracy and traction motor temperature.
- Cloud analytics learn a normal baseline for each lift and raise an anomaly alert when, for example, door closing time drifts by more than 15% or vibration rises above its baseline band.
- Every alert automatically creates a work order in the CMMS with the sensor evidence attached, so the technician arrives knowing which component to inspect.
- Spare door operators, door rollers and lock contacts are held on site because door faults dominate.
- The maintenance contract moved from monthly visits to an availability-based contract with a 99.5% availability target and shared savings.

4. Alert triage

In the first six weeks the system produced too many alerts and technicians began to ignore them. Thresholds were re-tuned for each lift using technician feedback on every alert ('confirmed fault' or 'false alarm'), which cut false alarms by two thirds.

5. Results (FY2024)

KPI	FY2023	FY2024
Breakdowns	58	31
Entrapments	7	2
Mean downtime per incident (hours)	5.2	2.1
Lift availability	98.7%	99.6%
Maintenance cost	baseline	+4% (sensors and subscription)

6. Lessons learned

- Monitor the doors first: they cause most failures and are the cheapest part to instrument.
- Availability-based contracts align the contractor's incentives with the occupant experience.
- Alerts need an owner and a feedback loop; untuned thresholds create alarm fatigue.
- Condition data is most valuable wherever lift downtime has a high human cost.